

Total time : 2 hour

Total Marks : 30

Instructions

1. Calculators are allowed.
2. Only black, blue pen and pencils (for diagram) are allowed.
3. In absence of Name and/or Roll Number, we may not award any marks to you.
4. If Name and/or Roll Number is not written till 20 minutes after the exam has started, instructor/invigilator has the right to take away the sheets.
5. Students are suggested to avoid any water or washroom break in this 2-hour exam.
6. No one is allowed to enter the room after 15 minutes of commencement of the exam. You are only allowed to leave and submit the exam after 30 minutes

1) Find basis and dimension of subspaces  $S$  of  $\mathbb{R}^3$  given

$$S = \{(x_1, x_2, x_3) \mid x_1 + x_2 - x_3 = 0\}$$

2

2) Comment on the linear transformation  $(T: \mathbb{R}^2 \rightarrow \mathbb{R}^3)$  such that

$$T(x_1, x_2) = (x_1 + x_2 + 1, 2x_1 - 3x_2, 9x_1 + 7x_2)$$

2

3) Define Manhattan and Euclidean norm for  $\mathbb{R}^n$

1

4) How to write matrix  $A$  as a eigen-decomposition form. Further, demonstrate how one can use this form to estimate  $A^n$ .

2

5) Write how one can write a matrix  $A$  using Singular Value Decomposition (SVD). If you have a matrix  $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$ , how to get its SVD. Solve only till the eigenvalues. No need to go beyond that

3

6) Write the formula for right pseudo inverse of  $A$

1

7) How regularization is useful for polynomial regression

1

8) Comment on following L.P.P., one can use graphical method

3

$$\text{Maximize } Z = 2x_1 + 3x_2$$

$$\text{Subject to } x_1 - x_2 \leq 2; x_1 + x_2 \geq 4; x_1 > 0; x_2 > 0$$

9) Express the following problem in the standard form

2

$$\text{Minimize } Z = 5x_1 + 9x_2$$

Subject to  $2x_1 - x_2 - x_3 = -9$ ;  $4x_1 + 6x_2 + 2x_4 = 21$ ;  $x_1 - 9x_2 = 3$ ;  $x_1, x_3, x_4 \geq 0$

10) Maximize  $-2x_1 - 2x_2 - 4x_3$

4

Subject to  $2x_1 + 3x_2 + 5x_3 \geq 2$ ;  $-3x_1 - x_2 - 7x_3 \geq -3$ ;  $x_1 + 4x_2 + 6x_3 \leq 5$ ;

$x_1, x_2, x_3 \geq 0$

Solve using dual simplex method.

11) A company has three cement factories located in cities A, B, C which supply cement to four projects located in places 1, 2, 3, 4. Each factory can supply 6, 10 truckloads of cement daily, respectively and daily cement requirements of the projects are 7, 5, 4, 2 truckloads. The transportation cost per truckloads of cement (in thousands of Rs from each plant to each project site are as follows:

| Factories | Project |   |   |    |   |
|-----------|---------|---|---|----|---|
|           |         | 1 | 2 | 3  | 4 |
| A         |         | 2 | 3 | 11 | 7 |
| B         |         | 1 | 0 | 6  | 1 |
| C         |         | 5 | 8 | 15 | 9 |
|           |         | 7 | 5 | 4  | 2 |

6  
2  
10

Determine the optimal distribution for the company as to minimize the total transportation cost.

5

12) There is a test to diagnose a disease. Test correctly detects when a patient has that disease 90% of the time. Also, if person does not have the disease, it correctly indicates 99.9% of the time. Suppose it is known that 1 in every 10,000 individuals has that disease. Find the probability that a patient has disease, given the test indicates they does.

2

13) Explain what is Entropy and let say if S is a collection of some sample with 10 apples and 6 oranges. What will be Entropy of that sample.

2